

Buffer Overflow attack

- A **buffer** is a temporary area for data storage.
- A **buffer overflow attack** occurs when a program writes more data to a buffer than it can hold, causing the excess data to overwrite adjacent memory. This can lead to data corruption, program crashes, or even the execution of malicious code

Types of Buffer Overflow Attacks:

- **Stack-based Buffer Overflow:**

This happens when the overflow occurs on the stack. It can overwrite the return address and other control data, allowing the attacker to control the flow of execution.

- **Heap-based Buffer Overflow:**

This type of overflow happens in the heap. It can corrupt function pointers or other critical data structures used by the program.

- **Format String Vulnerabilities:**

A type of overflow where an attacker can manipulate format strings (e.g., %s or %x) to leak information or overwrite memory.